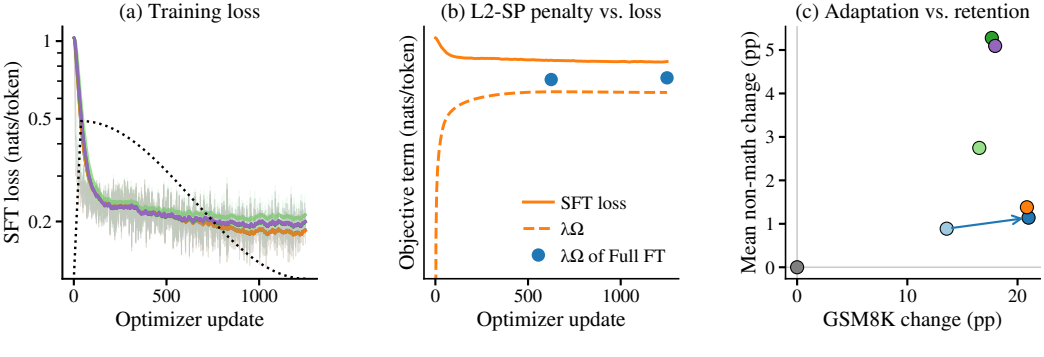

(a) Training loss
SFT loss (nats/token)
1
0.5
0.2
0
500
1000
Optimizer update
(b) L2-SP penalty vs. loss
Objective term (nats/token)
SFT loss
λΩ
λΩ of Full FT
0
500
1000
Optimizer update
(c) Adaptation vs. retention
Mean non-math change (pp)
5
4
3
2
1
0
0
10
20
GSM8K change (pp)
Baseline
Full FT
LoRA-4
OPLoRA-16
Full FT (mid)
Full FT + L2-SP
LoRA-16
LR / peak LR (a)